Medallion Architecture: A Practical Guide

Technical reference document for OCR-enabled RAG application testing

1. What Is Medallion Architecture?

Medallion Architecture is a way of organizing data in a lakehouse or data platform into three progressive layers: **Bronze**, **Silver**, and **Gold**. Each layer holds the same business information at a higher level of quality than the one before it. Think of it as a refinery: crude material enters at one end, is cleaned and checked along the way, and leaves as a finished product that analysts and dashboards can trust.

Rather than transforming data in a single risky step, the pipeline moves it through clearly defined stages. Each stage has one job, one owner, and one set of quality expectations, which makes the whole system easier to build, test, and troubleshoot.

2. The Three Layers

Bronze Layer (raw). The Bronze layer is the landing zone. Data arrives exactly as the source system produced it, including duplicates, gaps, and formatting quirks. Records are typically appended with metadata such as load time and source name. Because nothing is altered or discarded, Bronze acts as a permanent, replayable record of what was received.

Silver Layer (cleaned). The Silver layer improves quality. Pipelines remove duplicate records, handle missing values, standardize data types and formats, and validate each record against business rules. The result is a consistent, trustworthy dataset that is still detailed enough to support many different downstream uses.

Gold Layer (business-ready). The Gold layer is built for consumption. Validated Silver data is joined, aggregated, and shaped into tables that answer specific business questions, such as spending summaries per customer. Gold tables are optimized for reporting tools like Power BI and for fast, predictable queries.

3. How Data Moves Through the Pipeline

Raw customer transactions are first ingested and written to Bronze without modification. A cleaning job then reads Bronze, applies deduplication, missing-value handling, and validation rules, and writes the passing records to Silver. Records that fail validation are not silently dropped; they are routed to error handling and the issue is logged so it can be investigated and reprocessed. Finally, an aggregation job reads the validated Silver data, calculates business metrics, and publishes the Gold tables to the reporting layer. At every hop, data becomes cleaner, more structured, and closer to a form that decision makers can use.

4. Why This Architecture Is Useful

Medallion Architecture separates concerns, so a problem can be traced to a specific stage instead of an entire pipeline. Keeping raw data in Bronze allows any downstream layer to be rebuilt after a bug fix or a rule change. Incremental quality improvement supports clear data lineage, auditing, and governance, while Gold tables give analysts fast, consistent answers without needing to understand source-system details. Teams can also work on layers independently, which improves scalability and maintainability as data volumes and use cases grow.

Pipeline Flowchart and Layer Comparison

The flowchart below shows the end-to-end path of a customer transaction, including the branch taken when validation fails.

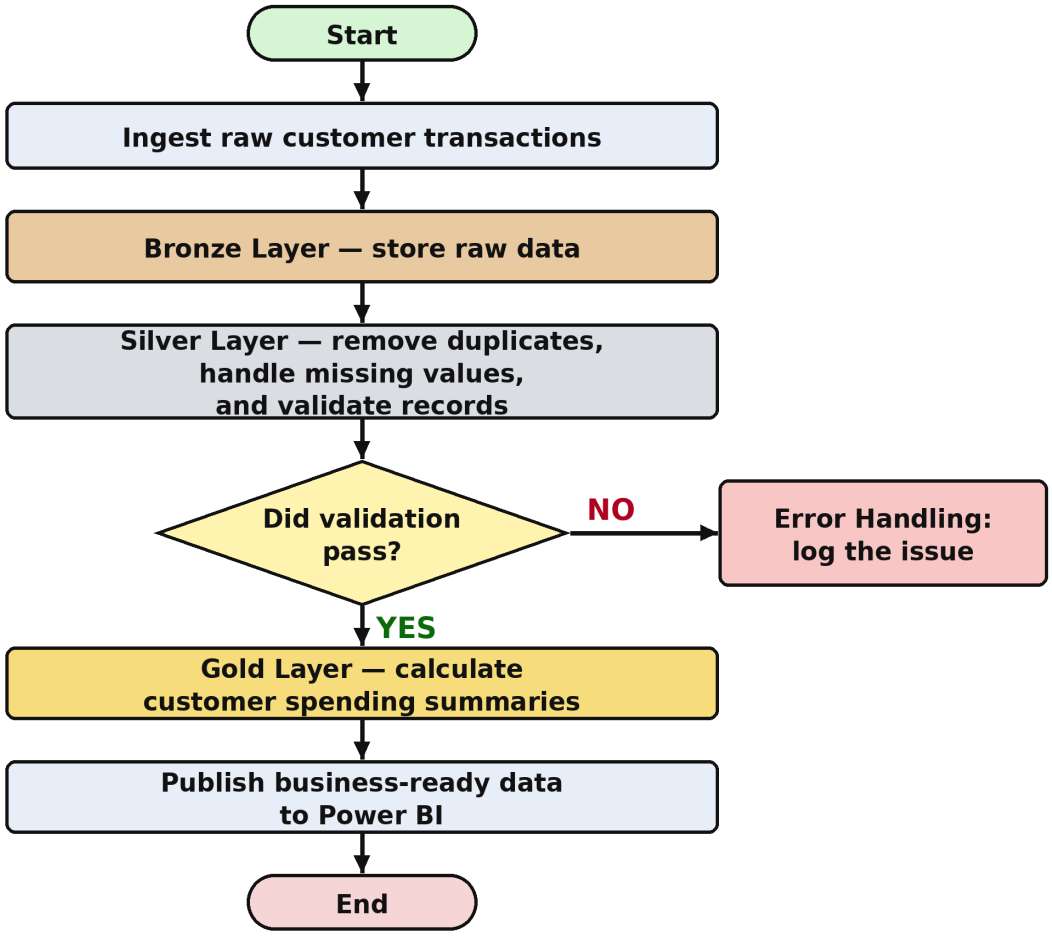


Figure 1. Medallion pipeline flowchart (embedded as an image).

Table 1. Layer comparison

Layer	Purpose	Example Data	Key Operation
Bronze	Preserve raw source data	`transactions_raw`	Append raw records
Silver	Improve data quality	`transactions_cleaned`	Deduplicate and validate
Gold	Support analytics	`customer_spending_summary`	Aggregate customer spending

Pipeline Facts Infographic

The infographic below is a single embedded image. Its contents are part of the picture rather than the page text.

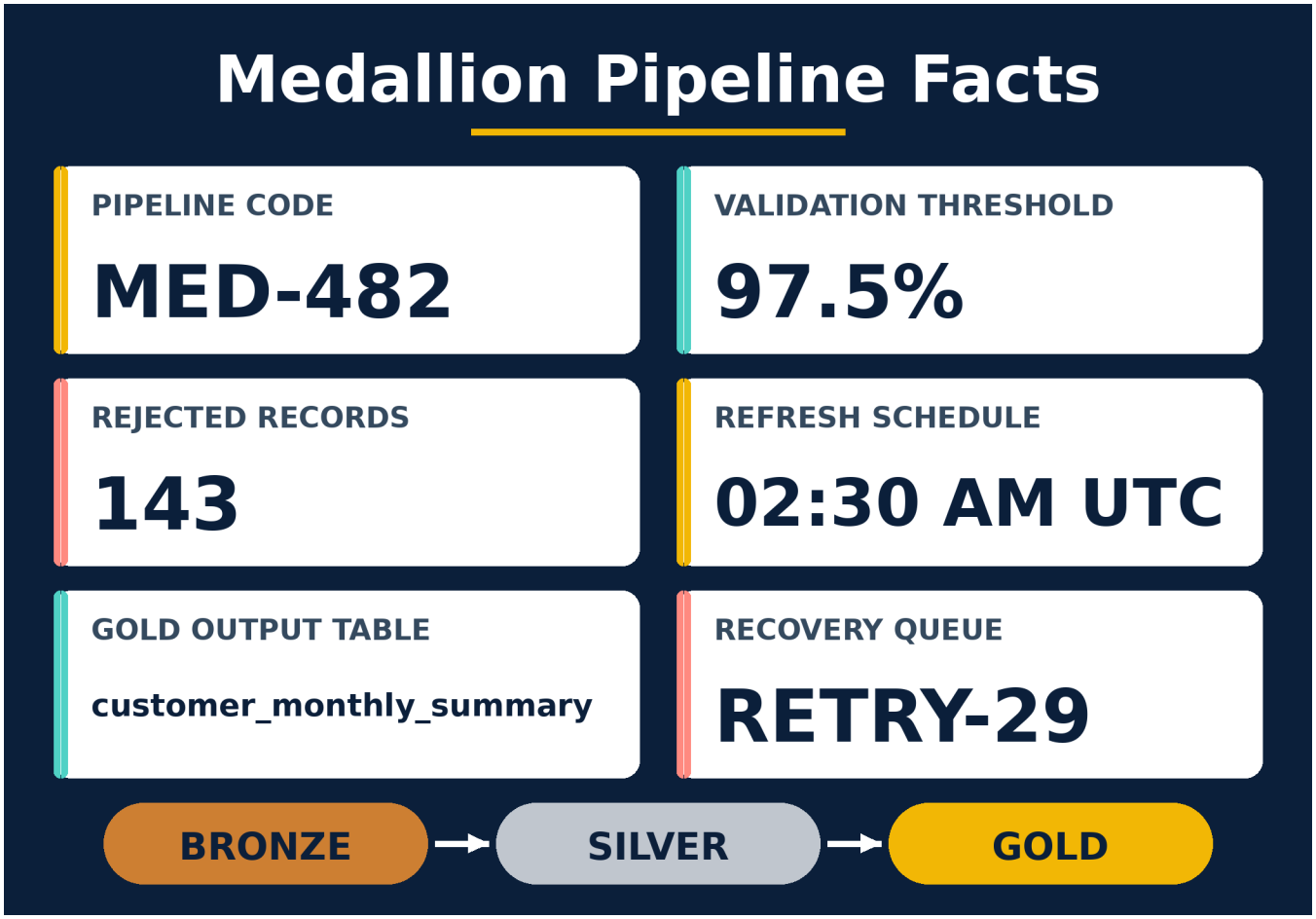


Figure 2. Medallion Pipeline Facts (embedded as an image).

What the Infographic Represents

The infographic works as a compact status dashboard for a single run of the medallion pipeline. It brings together six operational details in one place: the identifier of the pipeline, the quality bar that records must clear, the number of records that did not clear it, the final analytics table, how often that table is refreshed, and the queue where failed records wait for reprocessing. The Bronze, Silver, and Gold strip at the bottom reminds the reader of the order in which data is refined.

A reader can use this view to judge pipeline health at a glance: how strict validation is, how much data was set aside, where the business-ready output lands, and where to look when recovery is needed.